ARSPI-Net: An Event-Driven Reservoir–Graph Substrate for Embodied Affective EEG Perception

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***Abstract*—We present ARSPI-Net (Affective Reservoir–Spike Processing and Inference Network), a brain-inspired event-driven reservoir–graph substrate that transforms noisy affective electroencephalography (EEG) into a set of operationally distinct neural evidence streams and evaluates those streams as perceptual channels for a simulated embodied agent. The substrate is anchored in concrete neural mechanisms: a fixed leaky integrate-and-fire (LIF) spiking reservoir, a binned spike-count (BSC$_6$) event-driven code, an electrode-level temporal phase-locking (tPLV) graph, and a structure–function coupling readout κ. These produce four evidence streams: a spike-coded embedding E, dynamical descriptors D, graph-topological descriptors T, and a coupling block C. On 633 trial-averaged subject–condition observations from 211 adults under subject-grouped cross-validation across five seeds, a mechanism ablation with negative controls separates the streams: balanced accuracy is highest for the embedding-containing configurations (E+D+T, 0.481) and the band-power baseline (0.485), the standalone D, T, and C streams are lower but above the three-class chance level, and a shuffled-label control collapses to chance; an independent permutation–FDR protocol corroborates the ordering. The streams are not linear duplicates of one another. We characterise the streams under four perturbation families—temporal jitter, amplitude noise, channel dropout, and graph edge perturbation—and report their operating regimes rather than a single endpoint. Finally, an explicitly defined expected-free-energy controller uses the substrate as a perceptual estimator in a simulated closed-loop affective-control task with action-determined transition noise; against passive, random, pragmatic-only, epistemic-only, and perfect-perception oracle policies, the controller reaches and confirms a target affective state but shows no advantage over pragmatic control, an operating-regime result. Clinical labels are used only as exploratory contextual validation and are bounded by a false-discovery-rate result; no diagnostic claim is made. All results are restricted to the measured SHAPE ERP regime and are not a universal claim about EEG graph neural networks.**

***Index Terms*—EEG, neuromorphic computing, spiking neural networks, liquid state machines, reservoir computing, neural coding, predictive coding, sequential evidence accumulation, embodied AI, perception–action loops, structure–function coupling, layer ablation.**

I. Introduction

EEG offers millisecond-scale access to neural activity, but scalp recordings are noisy, indirect, and subject-dependent. For an embodied agent that must perceive and act on affective state, the central engineering problem is the extraction of usable neural evidence from this partially observed, nonstationary signal. We treat the affective event-related potential (ERP) as the output of a biological dynamical system and ask how an event-driven, brain-inspired substrate can transform it into evidence streams whose reliability can be measured under perturbation and used to drive perception under uncertainty. This is a neuromorphic signal-processing question, not a question of psychology or diagnosis. A single endpoint classifier is useful but blunt: it cannot tell us which part of a staged substrate carries the signal, how that signal degrades under realistic measurement perturbations, or whether two intermediate representations measure the same thing under different names.

Recent IEEE biomedical work sets a clear bar. Compact convolutional baselines such as EEGNet [6], EEG graph networks [8], [10], spiking–graph hybrids [9], and multimodal attention models [11], [12] are defended through cross-subject protocols, ablations, and module-level metrics rather than one accuracy number. The present manuscript follows that practice for ARSPI-Net (Affective Reservoir-Spike Processing and Inference Network), a staged neuromorphic architecture for clinical affective EEG.

ARSPI-Net produces four feature families per observation: a reservoir spike embedding E, dynamical descriptors D, electrode-level topological descriptors T, and structure–function coupling descriptors C. Earlier ARSPI-Net work characterised subject-covariance geometry and a four-level interpretability taxonomy. The question here is more architectural. Do E, D, T, and C behave as different things, or are they redundant relabellings of the same thing?

Operational distinctness is one of the evaluation criteria we apply to the substrate, alongside perturbation robustness and embodied-control utility. It is a measurement-oriented test of whether the streams are empirically non-equivalent under a fixed protocol—differing in predictive sufficiency, incremental utility, label-specific sensitivity, or representational redundancy—rather than redundant relabellings of one response. It does not assume the streams track independent biological mechanisms; it asks whether they behave as different signal-processing views of one EEG response.

The audit draws on three properties of this study. The data are a 211-subject affective EEG cohort with five overlapping psychiatric labels and per-subject comorbidity counts, an unusually clinically rich resource for affective EEG research. The architecture decomposes the staged pipeline into four blocks, each tied to a named neural mechanism rather than treated as a black-box feature extractor. The audit protocol combines subject-respecting permutation inference, FDR correction, re-

Inferential results use subject-respecting permutation tests at $n_{\mathrm{perm}} = 200$. Clinical contrasts are reported with Benjamini–Hochberg correction across the 30 diagnosis $\times$ configuration cells; no clinical contrast survived correction at $\alpha = 0.05$, so clinical results are presented as weak-signal sensitivity, not biomarker validation. This manuscript is a preprint and has not been certified by peer review.

dundancy under centered-kernel alignment [17], an electrode-permutation null on the coupling readout, and a sequential-decision simulation that places the four blocks in an embodied perception loop. The simulation is positioned as a precursor to active inference [29]–[31], not as an instantiation: it is closed-loop on the belief side, open-loop on the environment side.

This paper makes four bounded technical contributions. First, it specifies an event-driven reservoir–graph substrate in which each evidence stream is anchored to a named neural mechanism—LIF spiking, BSC_6 event-driven coding, tPLV connectivity, and κ structure–function coupling—with an explicit mathematical observable for each. Second, it establishes the operational distinctness of the streams through a mechanism ablation (A0–A9) with negative controls, distinguishing predictive utility (whether a stream raises endpoint accuracy) from measurement utility (whether a stream exposes structure the others do not). Third, it quantifies the perturbation robustness of the streams under temporal jitter, amplitude noise, channel dropout, and graph edge perturbation, reporting operating regimes rather than a single endpoint. Fourth, it evaluates the streams as perceptual channels for a simulated embodied affective-control loop driven by an explicitly defined expected-free-energy controller, compared against passive, random, pragmatic-only, epistemic-only, and oracle baselines. Throughout, dataset provenance and quality control are reported as verifiable facts, clinical labels are used only as exploratory validation structure bounded by a false-discovery-rate result, and outputs are privacy-preserving and reproducible.

II. Related Work

Brain-inspired computing treats biophysical neuron and synapse models as a computational substrate rather than an analogy. The leaky integrate-and-fire neuron is an idealised reduction of the Hodgkin–Huxley membrane [4]; liquid state machines [1] and echo state networks [2] use a fixed recurrent reservoir as a high-dimensional projection for simple readouts [3], and event-driven substrates such as Loihi [34] and SpiNNaker [35] run these primitives natively. Graph neural networks model EEG as electrode graphs but diverge in node features, graph construction, and readout [8], and spiking–graph hybrids add event-based temporal computation [9]. ARSPI-Net departs from end-to-end EEG classifiers by treating the reservoir as a fixed nonlinear measurement operator whose outputs feed several downstream descriptors rather than one classification head.

This paper adopts the standard EEG-evaluation practice of cross-subject protocols, controlled ablation, and claim-bounded clinical discussion [6], [7], [9], [11], and additionally evaluates the architecture as competing perceptual policies for a simulated embodied agent, closer to active-inference accounts of perception–action [29]–[31]. Relative to prior ARSPI-Net work—which characterised subject/condition geometry and an interpretability taxonomy—the question here is whether the four feature families behave as operationally distinct measurement layers and as perceptual channels under perturbation and embodied control.

III. ARSPI-Net Layer Formulation

A. Input Representation

Let

$$X_s^{(c)} \in \mathbb{R}^{N_{ch} \times T}, \qquad N_{ch} = 34, \tag{1}$$

denote the preprocessed event-related EEG response for subject s under condition $c \in \{\text{negative}, \text{neutral}, \text{pleasant}\}$. The affective task uses subject-condition observations directly. Clinical-label analyses average each feature family over the available conditions per subject.

B. Reservoir Dynamics and Spike Embedding

Each channel is fed independently into a fixed LIF reservoir whose discrete-time update is

$$\mathbf{u}_{t+1} = \beta \mathbf{u}_t + W_{in} x_t + W_{rec} \mathbf{z}_t - \mathbf{r}_t, \tag{2}$$

with spike emission

$$\mathbf{z}_{t+1} = H(\mathbf{u}_{t+1} - \theta), \tag{3}$$

where $H(\cdot)$ is the Heaviside step function, $\beta = 0.05$ is the membrane decay, $\theta = 0.5$ is the threshold, W_{in} and W_{rec} are Xavier-initialised input and recurrent matrices, and the reservoir size is $N_{\text{res}} = 256$. The recurrent matrix is rescaled so that its largest eigenvalue magnitude is 0.9. Spike counts are pooled into $n_{\text{bins}} = 6$ non-overlapping windows over the response interval $t \in [10, 70]$ to form the BSC_6 feature, and the per-channel BSC_6 vector is PCA-compressed to 64 components. The full embedding for a subject-condition observation is

$$E_s^{(c)} = [h_1 \| h_2 \| \cdots \| h_{34}] \in \mathbb{R}^{2176}, \tag{4}$$

where $h_v \in \mathbb{R}^{64}$ is the channel-level embedding.

C. Dynamical, Topological, and Coupling Blocks

The dynamical block D contains reservoir-response descriptors per channel. The topological block T summarises electrode-level connectivity as a temporal phase-locking matrix and reduces it to two per-electrode summaries (mean strength and clustering). For per-channel matrices $D_s \in \mathbb{R}^{34 \times p}$ and $T_s \in \mathbb{R}^{34 \times q}$, the rank-correlation coupling matrix is

$$K_s = \operatorname{corr}_\rho(D_s, T_s) \in \mathbb{R}^{p \times q} \tag{5}$$

and the coupling magnitude is

$$\kappa_s = \frac{\|K_s\|_F}{\sqrt{pq}}. \tag{6}$$

The block C contains κ_s alongside signed strength and clustering coupling summaries.

D. Operational Distinctness Criteria

For families $L_i, L_j \in \{E, D, T, C\}$ and target Y, the performance contrast is

$$\Delta_{ij}(Y) = \mathcal{M}(f(L_i), Y) - \mathcal{M}(f(L_j), Y), \tag{7}$$

with $\mathcal{M}$ either balanced accuracy or ROC-AUC and f a fixed L2-regularised logistic readout. The incremental utility relative to a baseline L_b is

$$\Gamma_i(Y|L_b) = \mathcal{M}(f([L_b \| L_i]), Y) - \mathcal{M}(f(L_b), Y). \tag{8}$$

Redundancy is assessed using linear centered-kernel alignment,

$$\mathrm{CKA}(X, Y) = \frac{\|X_c^\top Y_c\|_F^2}{\|X_c^\top X_c\|_F \|Y_c^\top Y_c\|_F}, \tag{9}$$

with X_c and Y_c column-centered [17].

IV. Brain-Inspired Mechanisms and Embodied Interpretation

The blocks of ARSPI-Net each have a named neural correlate. This section states the correlate, the implementation choice, and what each choice rules out, so the reader can treat the architecture as a sequence of physiologically motivated transformations rather than a stack of feature extractors. Table I summarises the mapping from each neural principle to its ARSPI-Net implementation, the computational object it produces, the observable it exposes, and its relevance to an embodied agent.

A. LIF Spiking Reservoir as a Recurrent Cortical Substrate

The reservoir realised by (2)–(3) is a fixed leaky integrate-and-fire network. The neuron model is the canonical biophysical reduction of the Hodgkin–Huxley axon [4]: a sub-threshold membrane that integrates synaptic drive, decays toward rest, fires when the integrated potential crosses threshold, and resets. With $\beta = 0.05$ the decay constant places the reservoir in the temporal regime that liquid state machines use for transient projection of input streams [1]. The recurrent matrix is held at spectral radius 0.9, just below the marginal-stability boundary at which a recurrent network's transient response becomes long enough to support readout while remaining contractive on average [2], [5]. The reservoir is not trained. The classifier on the readout side is.

The fixed-reservoir choice is deliberate. Cortical microcircuits do not appear to learn their recurrent connectivity through end-to-end gradient descent, and reservoir computing is one of the few tractable models that respects this [3]. Holding the reservoir fixed also forces every measurement and inference claim to come from properties of the recurrent dynamics themselves rather than from a classifier learning to compensate for a poorly chosen substrate. The price is that we cannot, within this paper, attribute clinical signal to plastic synaptic adaptation; on-line plasticity on the readout is left to future work.

B. BSC_6 as an Event-Driven Code

The embedding E in (4) is built from binned spike counts over $n_{\text{bins}} = 6$ windows of the post-stimulus interval $t \in [10, 70]$. The choice rests on three properties. First, BSC_6 is event-driven. The representation is read off spikes, not membrane traces, and the classifier never touches a continuous neuronal state. Second, the encoding window starts a small distance after stimulus onset and ends well before the stimulus offset, covering the temporal interval where ERP components such as P200, N200, and the early portion of P300 are expressed. The bins are coarse enough to be robust to small temporal jitter but fine enough to preserve coarse temporal structure within the response. Third, the format is hardware-friendly: spike counts over fixed time windows are precisely the operation that on-chip event counters provide on Loihi [34] and SpiNNaker [35], and binned spike-count classifiers have been shown to support unsupervised digit recognition [33]. PCA compression to 64 per channel gives a fixed-length representation that the downstream readout consumes without retraining the reservoir.

C. tPLV Connectivity as Cortical Network Organisation

The topological block T is derived from per-observation 34×34 temporal phase-locking value matrices [23]. PLV measures inter-electrode phase consistency across trials and is one of the standard tools in EEG functional connectivity. Computed within the event-related response window, tPLV captures the connectivity that is in force during the task rather than during rest, mitigating the stationarity concerns that follow from full-trace PLV estimates. The matrix is reduced to two per-electrode summaries, mean strength and clustering. Both belong to the established graph-theoretic vocabulary of brain network analysis [24], and both have a direct biological reading: strength tracks how synchronised an electrode is with the rest of the head, while clustering tracks whether its synchronised neighbours are themselves synchronised. The reduction loses information by design. We trade pairwise resolution for a per-channel descriptor compatible with the embedding shape used by the other blocks.

D. Structure–Function Coupling κ

The coupling readout (6) measures whether a subject's per-electrode dynamical signature (D) and topological signature (T) co-vary across electrodes. It is a single non-negative scalar bounded above by 1, computed per subject-condition pair, and admits an electrode-permutation null. Structure–function coupling is a recurring quantity in network neuroscience, where it indexes the degree to which functional patterns line up with the structural backbone [25]–[27]. The use here is closer to a within-subject signature than to a population-level claim: κ asks whether one subject's local electrode dynamics and the same subject's electrode-level network position are coordinated, and the permutation null asks whether the coordination is more than would arise from chance electrode labelling. The block C embeds κ alongside signed strength and clustering coupling.

TABLE I
NEURAL MECHANISMS IMPLEMENTED BY ARSPI-NET, EACH ANCHORED TO A CONCRETE COMPUTATIONAL OBJECT AND A MEASURED OBSERVABLE.

Neural principle	Implementation	Computational object	Measured observable	Relevance to embodied AI
Event-driven coding	BSC_6 binning	temporal-bin spike-count vector	bin evidence	sparse perceptual observation
Recurrent cortical dynamics	LIF reservoir	reservoir state trajectory	embedding E	nonlinear temporal transformation
Temporal state statistics	descriptor block D	trajectory descriptors	variance, autocorrelation, complexity	dynamical evidence stream
Functional connectivity	tPLV graph	adjacency / graph operator	topology descriptors T	spatial dependency
Structure-function coupling	κ	coupling readout	shuffle-null significance	systems-level coordination
Embodied perception	posterior update + policy	belief state	entropy, success, steps	closed-loop perceptual utility

E. Sequential Evidence Accumulation

The four streams are interpreted as operationally distinct sub-systems available to an embodied agent (Section VI-D). The agent maintains a Dirichlet posterior $\mathrm{Dir}(\boldsymbol{\alpha}_t)$ over a held-out subject's affective response; at each step it picks one of the four feature-stream policies and updates the posterior under the soft observation

$$\boldsymbol{\alpha}_{t+1} = \boldsymbol{\alpha}_t + \mathbf{p}_{\theta_a}(x_t), \tag{10}$$

where $\mathbf{p}_{\theta_a}(x_t)$ is the policy-a classifier's class-probability vector for x_t, and expected information gain is the KL divergence of the predictive distribution across the update. This is a sequential-decision counterpart to the offline ablation—which stream most rapidly reduces an agent's posterior uncertainty—closed-loop on the belief side and open-loop on the environment side, a precursor to active inference [29], [30].

V. EXPERIMENTAL PROTOCOL

A. Data, Splitting, and Privacy

The analysis uses 633 subject-condition observations from 211 community-recruited adults, exactly balanced across the three affective conditions (negative, neutral, pleasant). Clinical-label analyses use 206 subjects with available SUD, MDD, PTSD, GAD, and ADHD labels. Comorbidity is the rule rather than the exception in this cohort: most subjects carry more than one label, and the analyses below use comorbidity counts as a covariate where appropriate. All released artifacts use hashed subject identifiers. Raw EEG, clinical metadata, and subject-level features are restricted and not publicly released.

Table II summarises the analysis protocol.

B. Architecture Overview

Fig. 1 shows the staged ARSPI-Net architecture.

C. Feature Blocks and Diagnostics

Table III lists the evaluated blocks. The embedding E has shape $(633, 2176)$ with full standardised rank 632 and no constant columns.

D. Ablation Matrices

Affective classification used the A0–A9 configurations. The classifier is L2-regularised logistic regression with train-fold-only standardisation under a 10-fold StratifiedGroupKFold subject separation. Clinical-label sensitivity used the C1–C6 configurations at the subject level with the same logistic readout. Linear separability is the deliberate choice, because operational distinctness is a property of the feature spaces under a fixed read-out, not of an unconstrained nonlinear endpoint model. The optimisation is

$$\hat{\theta} = \arg\min_{\theta} \left[-\sum_{i=1}^{n} \log p_\theta(y_i|x_i) + \lambda \|\theta\|_2^2 \right] \tag{11}$$

and the primary metric is balanced accuracy

$$\mathrm{BA} = \frac{1}{K} \sum_{k=1}^{K} \frac{TP_k}{TP_k + FN_k} \tag{12}$$

with $K = 2$ for clinical-label sensitivity and $K = 3$ for affective classification.

VI. RESULTS

We organise the evidence as one argument over four pillars: (A) dataset provenance and the computational objects the substrate produces; (B) the reservoir–graph mechanism ablation; (C) perturbation robustness and operating regimes; and (D) the simulated embodied affective-control loop. The primary ablation is the multi-seed mechanism ablation of Section VI-B; an independent ten-fold permutation–FDR protocol corroborates it, and that protocol together with the secondary clinical, redundancy, and comorbidity analyses is reported in the supplementary material.

A. Dataset Provenance and Observed Computational Objects

The data are trial-averaged event-related EEG from 211 community-recruited adults (subject 127 excluded for a recording anomaly), giving 633 subject–condition observations balanced across Negative, Neutral, and Pleasant. The evaluation uses subject-grouped cross-validation, so no subject appears in both training and test. Subject-level data are held in a restricted research environment, and only aggregate or hashed outputs are reported.

TABLE II
AFFECTIVE AND CLINICAL ANALYSIS PROTOCOL

Stage	Input	Operation	Output / Control
Data	211 subjects, 633 subject-condition EEG observations	Condition-level affective analysis; subject-level clinical analysis	Three affective classes; 206 subjects with clinical labels
Feature diagnostics	BandPower, E, D, T, C	Check finite values, ranks, zero fractions, constant columns	Embedding E nonzero with standardised rank 632; all blocks nondegenerate
Affective ablation	A0–A9 feature configurations	10-fold subject-grouped CV with train-fold-only scaling	Balanced accuracy, macro ROC-AUC, macro-F1, predictions, confusion matrices
Clinical sensitivity	C1–C6 subject-average features	Binary logistic readout per clinical label	Exploratory clinical-label sensitivity, not diagnostic validation

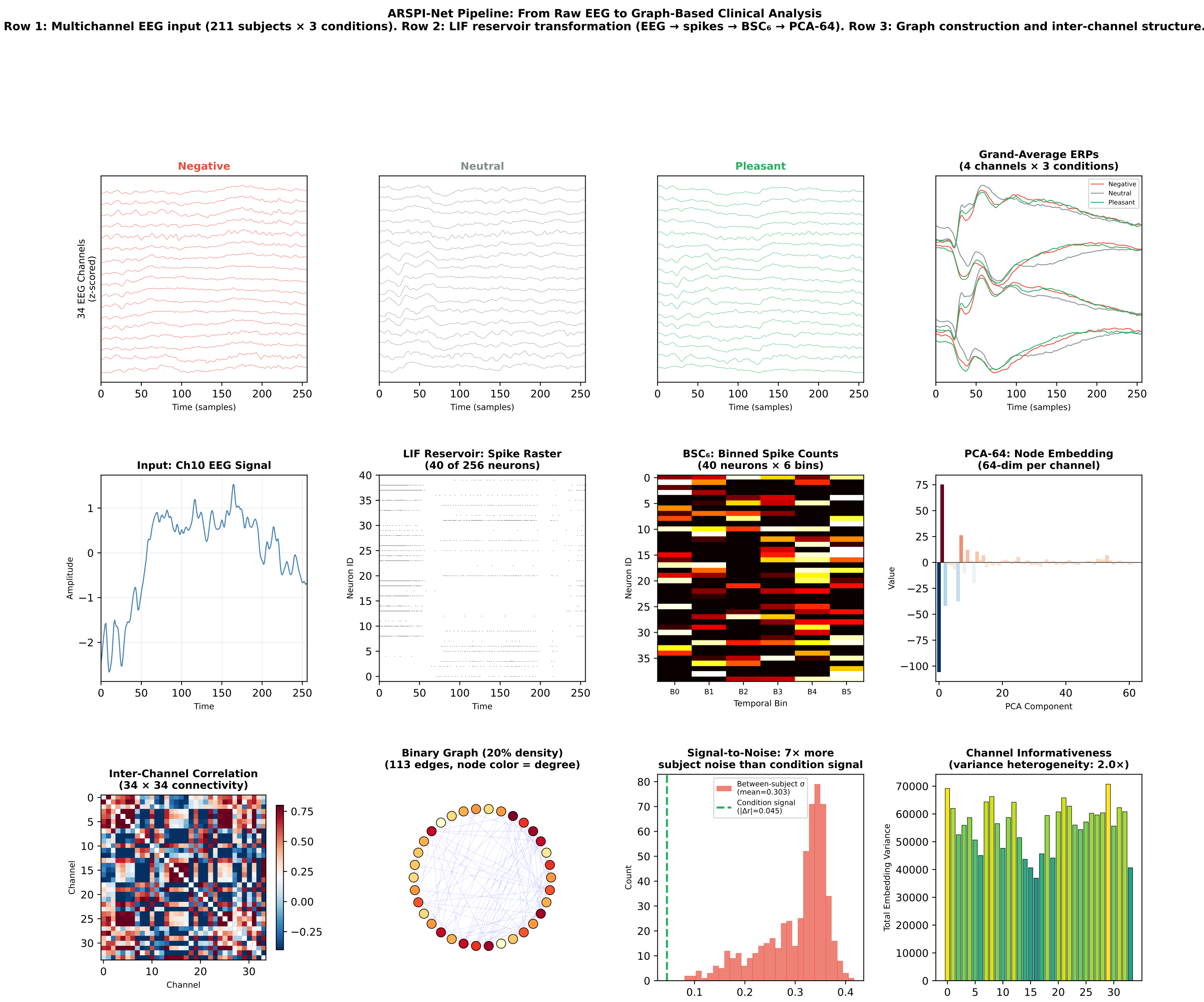

Fig. 1. ARSPI-Net pipeline overview: the staged event-driven reservoir–graph architecture, from the per-channel LIF reservoir and BSC_6 event-driven code to the embedding E, dynamical descriptors D, topological descriptors T, and coupling readout C evaluated in this work.

Fig. 2 surfaces the computational objects the substrate produces, for one exemplar per affective class: the LIF spike raster and membrane trajectory that generate the embedding E (panels a–b), the per-class 34×34 tPLV connectivity that defines the graph block T (panel c), and the BSC_6 projection that defines the event-driven code (panel d). Structure–function coupling is summarised by $\kappa = \|C\|_F/\sqrt{pq}$: across the 633 observations the mean is 0.250 (sd 0.115), and a 5,000-permutation electrode-shuffle null is rejected at $p < 0.05$ in 33.5% of observations, so a non-trivial subset of subjects show coordination above a randomised electrode labelling. Subject-mean κ is uncorrelated with per-subject classification

TABLE III
FEATURE FAMILIES USED IN THE OPERATIONAL-DISTINCTNESS ANALYSIS

Block	Dimension	Source	Operational role
BandPower	170	Spectral features	Conventional frequency-domain comparator
E	2176	BSC_6/PCA-64 reservoir embedding	Temporal spike representation
D	238	Reservoir trajectory descriptors	Dynamical response characterization
T	68	Graph-topological descriptors	Spatial organization across electrodes
C	3	Structure-function coupling	Alignment of dynamics and topology

accuracy ($\rho = 0.01$), behaving as a complementary systems-level readout rather than a proxy for predictive sufficiency.

B. Reservoir–Graph Mechanism Ablation

The primary ablation evaluates configurations A0–A9 under subject-grouped cross-validation across five seeds, with bootstrap intervals and two negative controls (Table IV, Fig. 3). Balanced accuracy is highest for the spectral band-power baseline (A0, 0.485) and the embedding-containing configurations (E+D+T 0.481, E+D+T+C 0.481, E+D 0.478, E 0.463); the standalone dynamical, topological, and coupling streams are lower (D 0.432, C 0.368, T 0.355) but remain above the three-class chance level of 0.333. The shuffled-label control collapses to chance (0.335), confirming the protocol. An independent ten-fold permutation–FDR protocol corroborates the ordering: eight of ten configurations exceed a subject-respecting permutation null, with standalone T and C the two that do not.

The reading is operational differentiation, not classifier superiority. The embedding carries most of the affective-classification signal, but D, T, and C are not linear duplicates of it—the largest pairwise centred-kernel alignment is $\mathrm{CKA}(E, D) = 0.31$—and they expose dynamical, topological, and coupling structure that the embedding alone does not. Clinical labels are used only as exploratory contextual validation: per-diagnosis balanced accuracies are near chance (0.49–0.53 across SUD, MDD, PTSD, GAD, ADHD) and no contrast survives Benjamini–Hochberg correction across the 30 diagnosis$\times$configuration cells; no diagnostic or clinical-biomarker claim is made. The clinical-sensitivity, redundancy, and comorbidity-adjusted analyses are reported in full in the supplementary material.

C. Perturbation Robustness and Operating Regimes

We characterise each evidence stream under four perturbation families—temporal jitter, amplitude noise, channel dropout, and graph edge perturbation—applied at the representation level across all ten configurations, with a bounded raw-signal diagnostic (on a bounded subset) that recomputes a reservoir embedding within fold. Fig. 4 reports the degradation profiles, which separate the streams by operating regime rather than by a single endpoint. Under additive amplitude noise the spectral band-power baseline is the most resilient, retaining 0.395 balanced accuracy at 5 dB SNR, whereas the E+D+T configuration falls toward chance (0.341, from a clean 0.481); the same ordering holds under channel dropout (0.408 versus 0.340 at 30% dropout). Graph edge perturbation, by contrast, leaves the embedding-dominated configurations essentially unchanged (E+D+T: 0.481 $\to$ 0.483 at 30%) while degrading the topological stream (T: 0.355 $\to$ 0.331). The measured regime is stream-specific—band-power and embedding descriptors are comparatively robust to signal-domain corruption and the graph descriptor is sensitive to graph-domain perturbation—and no claim of universal robustness is made. A supplemental nested evidence-routing analysis further bounds this result: an oracle over the streams shows large separable headroom, but label-free stream selection does not uniformly improve over the best fixed stream under the measured perturbation regime, supporting the interpretation of the substrate as making the operating regimes observable rather than as a universally dominant classifier.

A classical ERP-amplitude baseline (windowed $P1$–LPP features) provides a static discrimination anchor, reaching 67.2% balanced accuracy on the three-class task and exceeding ARSPI-Net under a static readout. The classical ERP baseline provides a strong static discrimination anchor; ARSPI-Net is evaluated as a reservoir–graph measurement substrate—decomposing the response into operationally distinct, perturbation-characterised evidence streams with simulated embodied utility—rather than as a replacement for ERP amplitude features in static classification. The full static comparison is reported in the supplementary material.

D. Simulated Embodied Affective-Control

We evaluate the substrate as the perceptual estimator for a simulated embodied agent. The agent maintains a categorical belief b over the affective state, updated by sequential Bayesian accumulation of the per-policy classifier likelihood (Section IV); the environment is action-determined with transition noise, $P(s' \mid a) = (1-\epsilon)\,\mathbf{1}[s'{=}a] + \epsilon\,\mathcal{U}$. The loop is a simulation over recorded observations and is not a physical embodiment.

Open-loop, the four streams act as competing observation channels: under sequential evidence accumulation their final posterior entropy ranks them E+D+T (fastest, 0.931 nats) $< E < D <$ random $< T$ (slowest, 1.075), the same ordering as the offline ablation, so offline sufficiency and perceptual-channel value agree.

Closed-loop, the agent must drive the environment to a target affective state and confirm it. An expected-free-energy (EFE) controller selects the action $a^\star = \arg\min_a \mathrm{EFE}(a)$ minimising

$$\mathrm{EFE}(a) = \underbrace{\mathrm{KL}\big(P(s' \mid a) \,\|\, C\big)}_{\text{risk}} + \underbrace{\mathbb{E}_{s' \sim P(\cdot|a)}\big[\hat{H}(s')\big]}_{\text{ambiguity}}, \quad (13)$$

where C is a preference distribution peaked on the target and $\hat{H}(s)$ is the expected observation-posterior entropy for

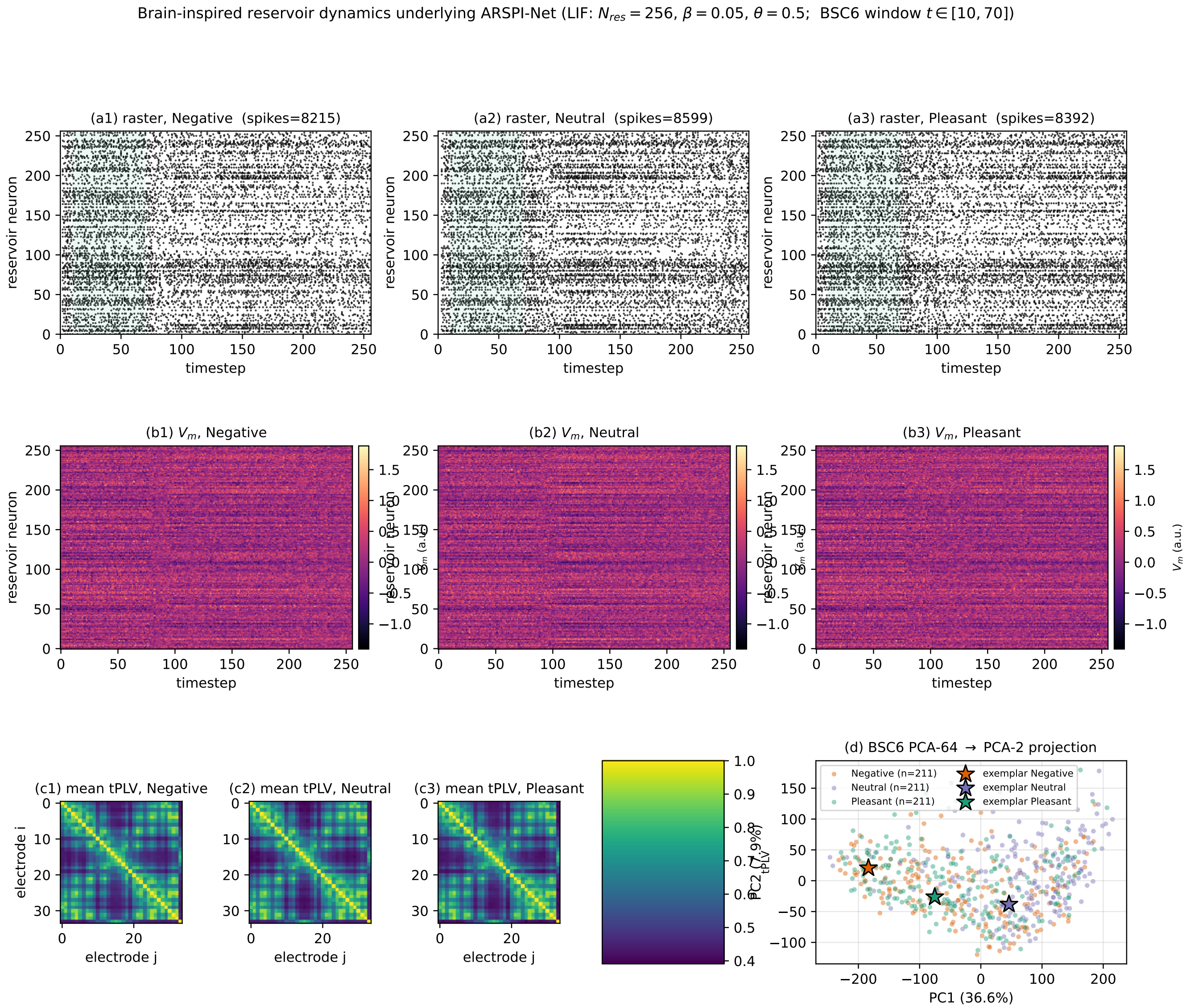

Fig. 2. Computational objects produced by the substrate. (a) LIF spike rasters for one exemplar per affective class on a representative channel; the BSC_6 encoding window $t \in [10, 70]$ is shaded. (b) Membrane-potential heatmaps for the same exemplars. (c) Per-class average 34×34 tPLV connectivity matrices across all 211 subjects. (d) BSC_6 PCA projection (PC1 vs PC2) coloured by class with the three exemplars marked.

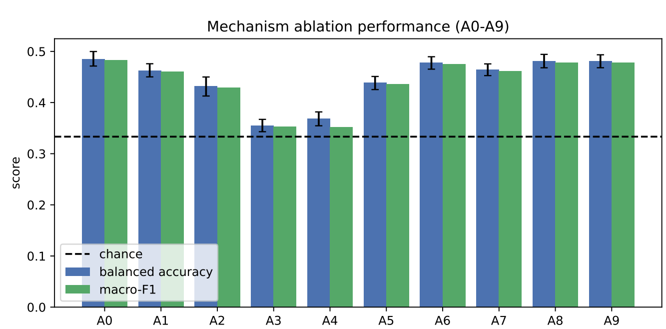

Fig. 3. Mechanism-ablation performance for configurations A0–A9. Balanced accuracy (with bootstrap intervals) and macro-F1 are shown against the three-class chance level. Bars summarise the measured cross-validated performance of each evidence stream and combination.

state s estimated on the training fold; the agent stops when $b(\text{target}) \geq 0.8$. It is compared against passive (belief-blind), random, pragmatic-only (risk term only), epistemic-only (ambiguity term only), and a perfect-perception oracle, over $1{,}500$ episodes per policy at each transition-noise level ϵ (Table V, Fig. 5). The oracle bounds success at 1.000. The EFE and pragmatic controllers reach and confirm the target (0.862 and 0.864 at $\epsilon = 0$; 0.805 and 0.808 at $\epsilon = 0.4$), while belief-blind control degrades fastest with transition noise ($1.000 \rightarrow 0.733$) and the random and epistemic-only baselines stay low. The EFE controller shows no advantage over pragmatic-only control: under an action-determined transition the epistemic term carries little marginal utility, and the gap to the oracle quantifies the cost of single-trial perceptual unreliability. This is an operating-regime result, not a controller-superiority claim.

TABLE IV
MECHANISM ABLATION UNDER SUBJECT-GROUPED CROSS-VALIDATION. BALANCED ACCURACY (BA) AND MACRO-F1 ARE REPORTED AS MEAN WITH A BOOTSTRAP 95% INTERVAL. CHANCE BA $= 0.333$. EACH ROW REPORTS THE FEATURE CONFIGURATION, THE NEURAL MECHANISM IT CARRIES, AND ITS FEATURE DIMENSION.

Cfg	Mechanism	Dim	BA	95% CI	macro-F1
A0	spectral band-power (non-neuromorphic baseline)	170	0.485	[0.471, 0.500]	0.483
A1	LIF reservoir spike embedding (E)	2176	0.463	[0.450, 0.476]	0.460
A2	reservoir dynamical descriptors (D)	238	0.432	[0.413, 0.450]	0.429
A3	tPLV graph topology descriptors (T)	68	0.355	[0.343, 0.367]	0.353
A4	structure-function coupling (C)	3	0.368	[0.355, 0.382]	0.352
A5	D+T (dynamics + topology)	306	0.439	[0.425, 0.451]	0.436
A6	E+D (embedding + dynamics)	2414	0.478	[0.465, 0.490]	0.475
A7	E+T (embedding + topology)	2244	0.464	[0.453, 0.476]	0.461
A8	E+D+T (full reservoir-graph substrate)	2482	0.481	[0.468, 0.494]	0.478
A9	E+D+T+C (substrate + coupling readout)	2485	0.481	[0.468, 0.493]	0.478
CTRL_shuffled_label	negative control	2482	0.335	[0.320, 0.351]	0.333
CTRL_shuffled_subject	negative control	2482	0.474	[0.461, 0.487]	0.471

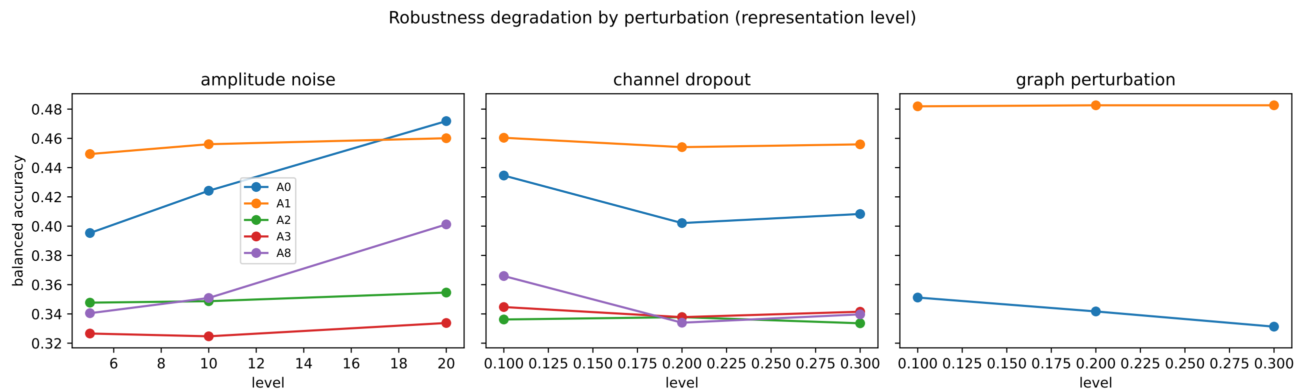

Fig. 4. Representation-level robustness degradation under amplitude noise, channel dropout, and graph edge perturbation for representative configurations. Curves report balanced accuracy as a function of perturbation level; they expose the operating regime of each evidence stream rather than a single endpoint.

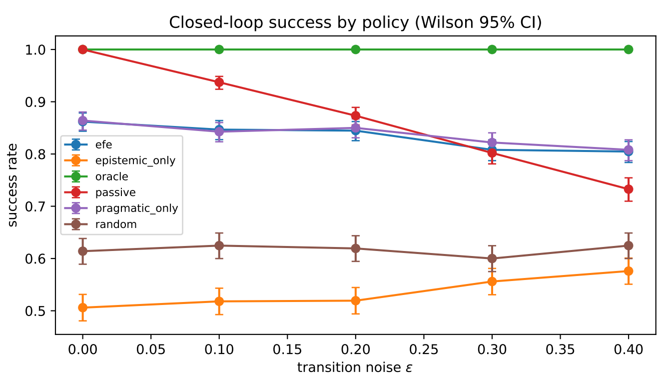

Fig. 5. Closed-loop success rate by policy and transition noise ϵ, with Wilson 95% intervals. The expected-free-energy and pragmatic controllers reach and confirm the target affective state and degrade gracefully; belief-blind, random, and epistemic-only baselines do not match them.

VII. DISCUSSION

The mechanism ablation establishes operational differentiation. Embedding-containing configurations carry most of the affective-classification signal, the standalone dynamical, topological, and coupling streams are weaker classifiers yet are not linear duplicates ($\mathrm{CKA}(E, D) = 0.31$) and expose structure the embedding does not, and a shuffled-label control collapses to chance. A block can be sub-threshold as a classifier and still operationally distinct as a measurement view. The brain-inspired claim rests on the named mechanisms of Section IV—LIF spiking, BSC_6 event-driven coding, tPLV connectivity, and κ coupling—rather than on analogy.

Perturbation robustness locates a stream-specific operating regime rather than a single ordering: the band-power and embedding descriptors are comparatively robust to signal-domain corruption (amplitude noise, channel dropout), while the topological descriptor is the component sensitive to graph-domain perturbation. Resource accounting in the supplemental material reports reservoir size, event rate, spike sparsity, and runtime estimates—the recurrent layer performs on the order of the measured spike-sparsity fraction of the dense-equivalent operation count, mappable onto a small number of event-driven neuro-cores [34], [35]—and these measurements support the event-driven substrate interpretation but do not constitute measured hardware-energy results.

The simulated embodied-control result establishes that the substrate functions as a usable perceptual estimator: an expected-free-energy controller reaches and confirms a target affective state and degrades gracefully with transition noise, bounded above by a perfect-perception oracle. It does not establish controller superiority—under an action-determined transition the epistemic term carries little marginal utility,

TABLE V
CLOSED-LOOP POLICY COMPARISON. SUCCESS RATE (WILSON 95% CI), MEAN STEPS, AND FINAL POSTERIOR ENTROPY BY TRANSITION NOISE ϵ.

Policy	ϵ	Success	Mean steps	Final $\mathcal{H}$
efe	0.0	0.862 [0.844, 0.878]	5.00	0.210
efe	0.1	0.847 [0.828, 0.864]	5.42	0.211
efe	0.2	0.845 [0.826, 0.862]	5.54	0.220
efe	0.3	0.808 [0.787, 0.827]	5.94	0.209
efe	0.4	0.805 [0.784, 0.824]	6.12	0.197
epistemic only	0.0	0.506 [0.481, 0.531]	8.86	0.122
epistemic only	0.1	0.518 [0.493, 0.543]	8.78	0.122
epistemic only	0.2	0.519 [0.494, 0.544]	8.88	0.141
epistemic only	0.3	0.556 [0.531, 0.581]	8.54	0.139
epistemic only	0.4	0.576 [0.551, 0.601]	8.44	0.152
oracle	0.0	1.000 [0.997, 1.000]	1.00	0.000
oracle	0.1	1.000 [0.997, 1.000]	1.07	0.000
oracle	0.2	1.000 [0.997, 1.000]	1.16	0.000
oracle	0.3	1.000 [0.997, 1.000]	1.23	0.000
oracle	0.4	1.000 [0.997, 1.000]	1.37	0.000
passive	0.0	1.000 [0.997, 1.000]	1.00	0.459
passive	0.1	0.937 [0.924, 0.949]	1.00	0.474
passive	0.2	0.873 [0.856, 0.889]	1.00	0.478
passive	0.3	0.802 [0.781, 0.821]	1.00	0.464
passive	0.4	0.733 [0.710, 0.754]	1.00	0.478
pragmatic only	0.0	0.864 [0.846, 0.880]	4.95	0.207
pragmatic only	0.1	0.843 [0.823, 0.860]	5.55	0.209
pragmatic only	0.2	0.850 [0.831, 0.867]	5.45	0.217
pragmatic only	0.3	0.822 [0.802, 0.841]	5.68	0.205
pragmatic only	0.4	0.808 [0.787, 0.827]	6.04	0.206
random	0.0	0.614 [0.589, 0.638]	8.17	0.186
random	0.1	0.625 [0.600, 0.649]	8.20	0.186
random	0.2	0.619 [0.595, 0.644]	8.25	0.186
random	0.3	0.600 [0.575, 0.625]	8.42	0.179
random	0.4	0.625 [0.600, 0.649]	8.10	0.184

so the expected-free-energy controller matches pragmatic control—and the oracle gap quantifies the cost of single-trial perceptual unreliability. The loop is a sequential-decision precursor to active inference [29], [30], closed-loop on the belief side and open-loop on the environment side, and is not a physical embodiment.

VIII. LIMITATIONS

The scope is bounded in five ways. The embodied evaluation is a closed-loop *simulation* over recorded EEG observations, not physical robot embodiment or live deployment. The clinical analysis is exploratory contextual validation only and is not diagnostic validation; no biomarker claim is made, and no clinical contrast survives Benjamini–Hochberg correction. Runtime and resource figures are structural; hardware energy is not measured and no low-power deployment is claimed. The dataset is a private, access-controlled clinical resource, which limits public benchmarking to approved-access conditions. All results are restricted to the measured SHAPE ERP regime and should not be read as universal claims about EEG graph neural networks. The static ERP-amplitude baseline outperforms ARSPI-Net on three-class accuracy; the contribution is the substrate's operationally distinct, perturbation-characterised evidence streams and their simulated embodied utility, not endpoint accuracy.

IX. CONCLUSION

ARSPI-Net is a brain-inspired event-driven reservoir–graph substrate that transforms affective EEG into operationally distinct evidence streams—a spike-coded embedding, dynamical descriptors, a phase-locking graph, and a structure–function coupling readout. A multi-seed mechanism ablation with negative controls separates the streams; a perturbation analysis reports their operating regimes; and a simulated embodied affective-control loop with an explicitly defined expected-free-energy controller evaluates them as perceptual channels. The streams behave as operationally distinct measurement views of one EEG response, and the substrate supports the brain-inspired-computing-for-embodied-AI framing on grounds of mechanism and of a simulated closed-loop demonstration, with clinical claims bounded and scope restricted to the measured regime.

DATA AND CODE AVAILABILITY

The analysis code and figure-generation scripts that produce the quantitative results are maintained in the project repository (link provided under approved-access and review conditions). Raw EEG, clinical metadata, and subject-level feature artifacts are maintained in a restricted research environment. The manuscript reports aggregate, deidentified experimental outputs and methodological details sufficient for reproduction under approved data-access conditions. The SHAPE Community dataset is access-controlled and available from the Laboratory for Clinical Affective Neuroscience at Stony Brook University, subject to approval and a data-use agreement. Aggregate, deidentified results are reported; restricted subject-level data are not publicly distributed and require approved access. Committed prediction-level artifacts use hashed subject identifiers.

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